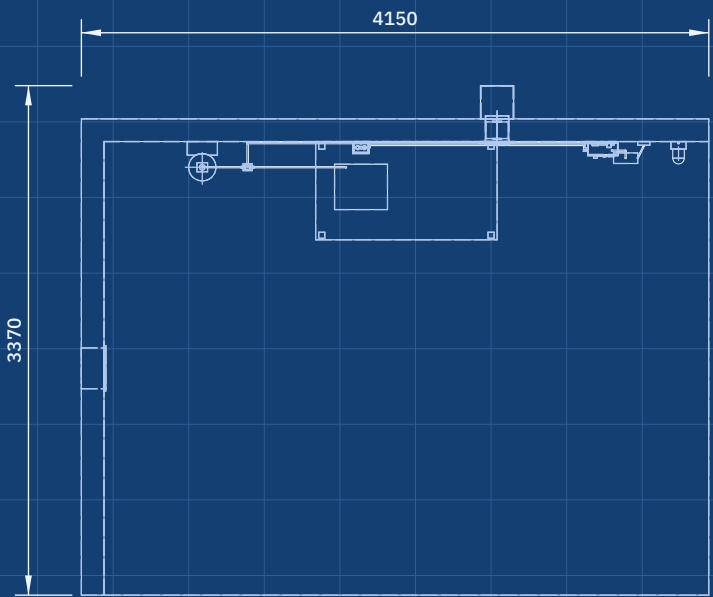


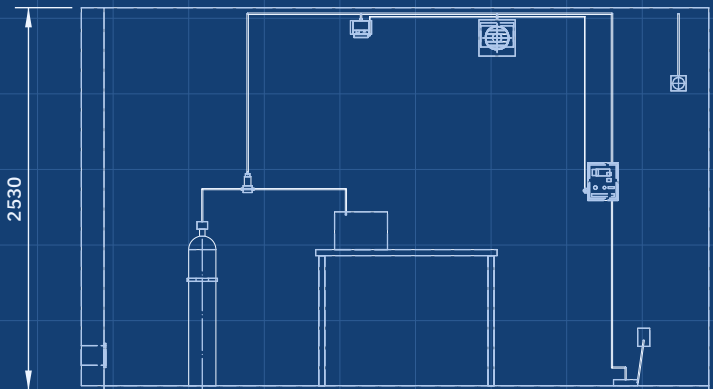
CONCEPT, NOT FOR FABRICATION

H2Guard · github.com/BoujeeEnjinia1701/h2guard



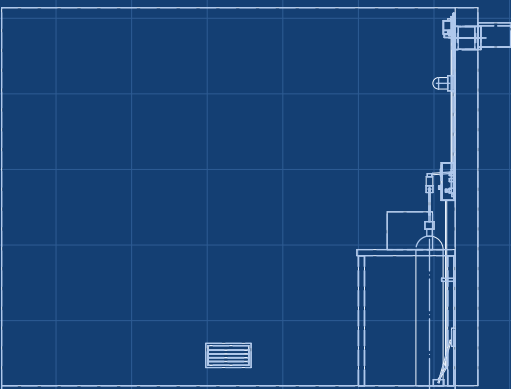
TOP VIEW

Scale 1:50; looking down (along -Z)



FRONT VIEW

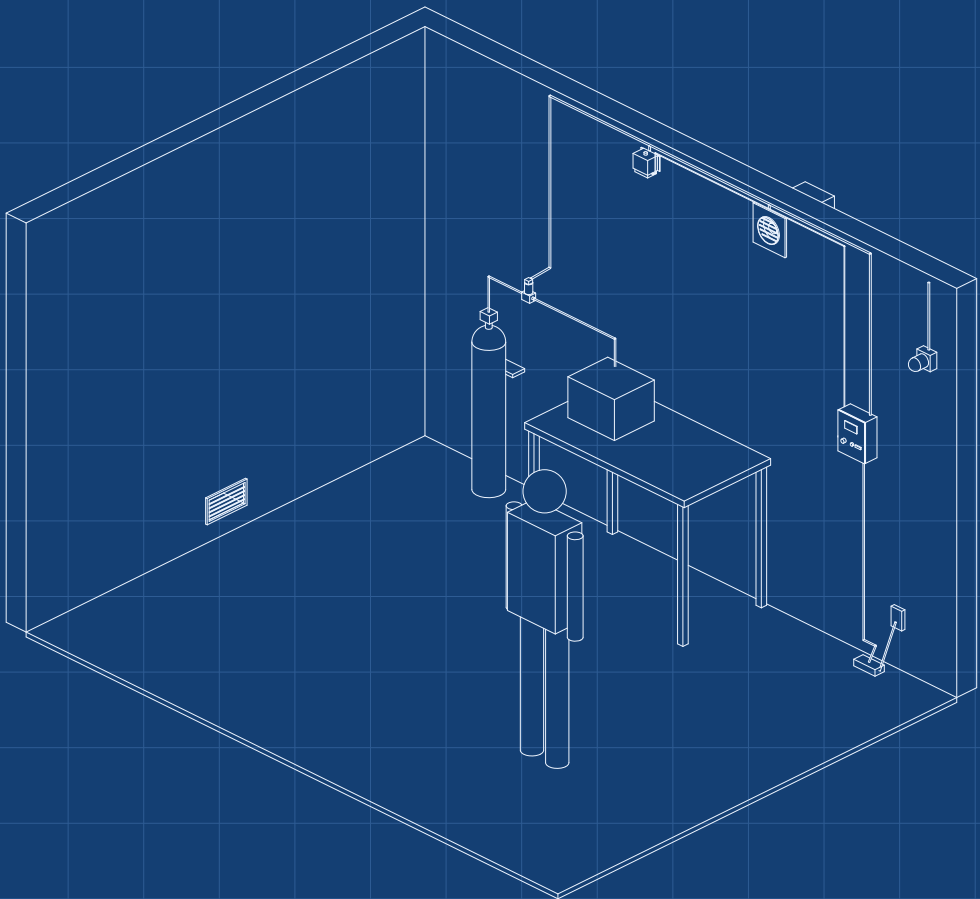
Scale 1:50; looking at the front (along +Y)



RIGHT VIEW

Scale 1:50; looking at the right side (along -X)

REV	DESCRIPTION	DATE	BY
P1	Concept sheet	2026-09-25	AC
P2	From the TRL 3 parametric model	2026-09-25	AC
P3	Key figures after DDR-002	2026-09-25	AC



ISOMETRIC VIEW

Not to scale; figure is a 1.75 m person; seen from the front right and above, about 30 deg elevation

KEY FIGURES

Warn at 10 % LFL (0.4 % vol H₂); trip at 25 % LFL (1.0 % vol)
Leak to valve closed 35.7 s with t₉₀ 30 s assumed (HGD-CAL-001)
Exhaust 150 m³/h continuous, 347 m³/h boost; room 5 % LFL
Fail-safe: NC valve, series hardware trip relay, 1 s watchdog
24 V DC build; mains only inside a certified supply
Warning held 5 min also closes the valve (DDR-002)
Parts \$264 (indicative); budget \$265

H2GUARD

github.com/BoujeeEnjinia1701/h2guard

Leak detector and interlock concept

DRAWING NO. HGD-DWG-010		REV P3	SHEET 1 of 1	SIZE ANSI B
SCALE 1:50	UNITS mm	PROJECTION 	DATE 2026-09-25	DRAWN Amish Chadha

MATERIAL / NOTES

Built from cad/src/model.py; room, bench and gas store are context. GA is HGD-DWG-001

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Generated from cad/src; do not edit by hand